

9.3(problems by hand)

q.3

1)

Learning rate

x^1	x^2	w_{11}^h	w_{21}^h	w_{11}^o	w_{21}^o	K	η
0.59	0.1	0.6	0.1	0.4	0.3	1	0.3

Net input $[I, J] = \sum w_i \cdot x_i + \theta_j$ output $(\theta_j) = \frac{1}{1 + e^{-h_j}}$

output = $\frac{1}{1 + e^{-0.364}} = 0.59$

$h_1 = w_{11} \cdot x_1 + w_{21} \cdot x_2 + \theta_{h_1}$

$= (0.6 \times 0.59) + (0.1 \times 0.1)$

$= (0.354) + (0.01) = 0.364$

Output: $y_1 = w_{11}^o \times \theta_{h_1} = 0.4 \times 0.59 = 0.236$

$y_2 = w_{21}^o \times \theta_{h_1} = 0.3 \times 0.59 = 0.177$

q3

2) Calculating Errors

hidden
node
error

$$O_i (1 - O_i) (\sum_k \text{err}_k w_{jk})$$

$$h_1 \text{ } \delta = 0.59 (1 - 0.59) ((0.6297 \times 0.9) + (0.5292 \times 0.3))$$

$$\delta = 0.59 (0.41) (0.72549) = \boxed{0.605}$$

output node

$$O_j (1 - O_j) (T_j - O_j)$$

~~$$y_1 = 0.6297 (1 - 0.6297)$$~~

$$y_1 \text{ } \delta = 0.6297 (1 - 0.6297) (1 - 0.6297) = \boxed{0.466}$$

$$y_2 \text{ } \delta = 0.5292 (1 - 0.5292) (1 - 0.5292) = \boxed{0.499}$$

Calculating the weight from the errors for each connection node

x_1

$$w_{11}^h = 0.9 \times 0.6 + 0.605 = \boxed{1.205} \quad w_{11}^h$$

x_2

$$w_{21}^h = 0.1 + 0.605 = \boxed{0.705} \quad w_{21}^h$$

y_1

$$w_{11}^o = 0.9 + 0.3 (0.59 \times 0.605) = 0.9 + 0.107 = \boxed{1.07} \quad w_{11}^o$$

y_2

$$w_{12}^o = 0.3 + 0.3 (0.605 \times 0.59) = 0.3 + 0.107 = \boxed{0.407}$$

z

$$0.3 + 0.107$$

$$= 0.407$$